

Basic Information

Experiment No.: 06

Date: 24-09-2026

Title: Java NavigableSet Interface using TreeSet

Objective / Aim

To implement the NavigableSet interface using "TreeSet" and demonstrate navigation, removal, and descending-order operations.

System Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM

Software

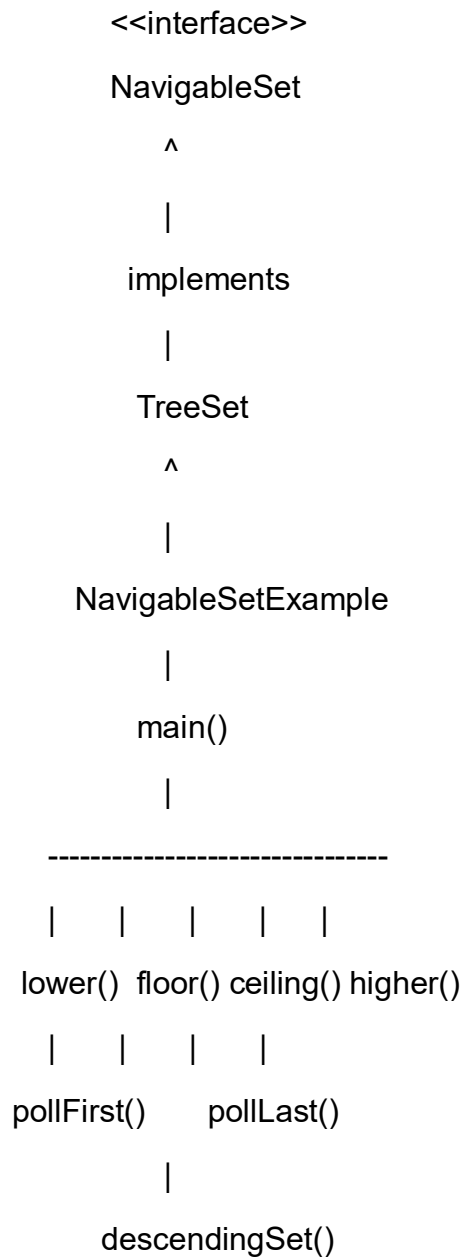
- Java JDK
 - Any Java IDE or Command Prompt
 - Operating System: Windows/Linux/macOS
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Algorithm / Logic Description

1. Create a "NavigableSet" using "TreeSet".
2. Add elements using "add()".
3. Find elements using "lower()", "floor()", "ceiling()", and "higher()".
4. Remove the first element using "pollFirst()".
5. Remove the last element using "pollLast()".

6. Display the remaining set.
 7. Display the set in descending order using "descendingSet()".
 8. End the program.
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UML Class Diagram



Source Code

```
package collectionframework;

import java.util.NavigableSet;
import java.util.TreeSet;

public class NavigableSetExample {
    public static void main(String[] args) {

        NavigableSet<Integer> set = new TreeSet<>();

        set.add(10);
        set.add(20);
        set.add(30);
        set.add(40);
        set.add(50);

        System.out.println("Set: " + set);

        // lower()
        System.out.println("Lower than 30: " + set.lower(30));

        // floor()
        System.out.println("Floor of 30: " + set.floor(30));

        // ceiling()
        System.out.println("Ceiling of 25: " + set.ceiling(25));

        // higher()
```

```
System.out.println("Higher than 30: " + set.higher(30));

// pollFirst()
System.out.println("Removed First: " + set.pollFirst());

// pollLast()
System.out.println("Removed Last: " + set.pollLast());

System.out.println("After polling: " + set);

// descendingSet()
System.out.println("Descending Set: " + set.descendingSet());
}
}
```

Compilation & Execution Commands

Save the file as:

NavigableSetExample.java

Since the program contains:

package collectionframework;

the file should be inside the "collectionframework" folder.

Compile:

```
javac NavigableSetExample.java
```

Run:

```
java collectionframework.NavigableSetExample
```

Output Screen

Set: [10, 20, 30, 40, 50]

Lower than 30: 20

Floor of 30: 30

Ceiling of 25: 30

Higher than 30: 40

Removed First: 10

Removed Last: 50

After polling: [20, 30, 40]

Descending Set: [40, 30, 20]

Result

The NavigableSet interface was successfully implemented using "TreeSet", and navigation, removal, and descending-order operations were successfully demonstrated.